

Visualization of Covid-19 Pandemic Data: An Analysis

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Abstract—This paper focuses on data visualization techniques and its efficiency. Data visualization refers to pictorial representation of data. Though it is easy to view and understand, the time is yet to come when this technique would see its full utilization. It has a lot of applications in data presentation as well as data exploration, pattern mining and predictive analysis. Its applications would be studied in relation to Covid-19 pandemic. The world is suffering from novel corona virus since December 2019. It has been more than two years and there seems no end to this pandemic. Data visualization techniques will be applied using R programming language, it has been widely adopted by statisticians to study and analyse data, so it will be used here as a tool to study data generated by corona virus outbreak in the last two years. However, the scope of the study is limited to data generated in India for number of confirmed cases, deaths and recovered cases.

Keywords— *Coronavirus, Covid-19, SARS-CoV-2, Data Visualization*

I. INTRODUCTION

Data Analytics is the pictorial or graphical representation of data. There are graphs, charts, maps etc that are used for this purpose. It is easy to make as well as easy to understand. With the growth of data, visualization[1] techniques have gained popularity but still it is not yet used to its optimal utility. A lot of time wrong conclusions are drawn from data presented in graphical form so it is necessary to understand the utility and applications of this powerful tool.

In the digital era, there is a lot of data generated every second, every minute, every day and so on, in other words big data. Big Data as the word itself suggests, it is humongous amount of data that is collected from a variety of sources. The data can be structured or unstructured. It is identified using the three V's:

- **Volume:** It refers to the huge amount of data collected from various sources like social media, traffic control systems and sensors, different websites, smart IoT devices etc.
- **Velocity:** It refers to the speed at which data is being collected. It can be near real time data that is generated by sensors and other IoT devices.

- **Variety:** It refers to various forms of data that is videos, texts, audios, images etc.

But there is no use opening a tap of water and letting the bucket overflow when there is no one to use the water. Similarly, here data is water which is being generated at massive rate. However, it is of no use if it is not being utilized for drawing conclusions, rectifying errors or taking any other action. Big data cannot be handled using traditional means, it is very time consuming and erroneous, for this various tool have been developed.

Data visualization is a technique which can be used to present big data and explore it for various alternative solutions to problems or predict future scope in any field using data generated in it. Before knowing the scope of anything it is necessary to know its roots, to grow more. The history of data visualization has not been a constant growth, it has various phases of its own before coming into the statistical world.

II. DATA VISUALIZATION

A. History of data visualisation

One of the most essential aspects of a developed country's economy is its healthcare system. The fear of COVID-19 has infected the entire world nowadays, and it is linked to the rapid growth of essential data, as well as the need to analyse the relationships and hierarchy between data.

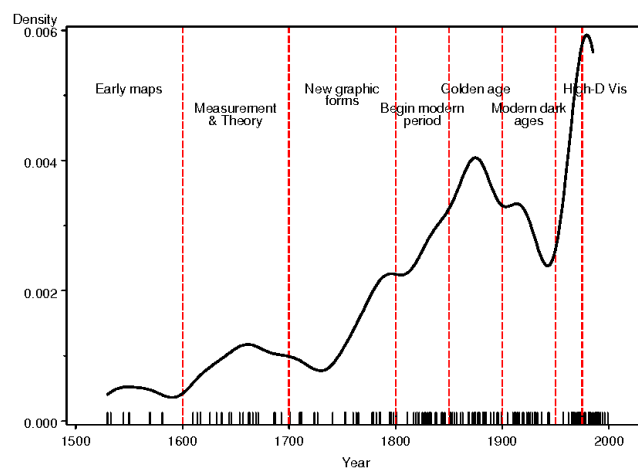


Fig 1. History of Data Visualization (Source: Mathew Friendly, 2006)

B. Big Data Visualization Methods

Many traditional data visualization techniques are frequently employed. Tables, histograms, scatter plots, line charts, bar charts, pie charts, area charts, flow charts, bubble charts, numerous data series or combinations of charts, time lines, Venn diagrams, data flow diagrams, and entity connection diagrams are just a few examples. Furthermore, other data visualization approaches have been utilized, however they are less well-known than the ones mentioned above. Parallel coordinates, treemap, cone tree, and semantic network are some of the additional ways.

C. Scope of data visualisation

We can no longer dispute the existence and significance of data anywhere. Whether you're a buyer or a seller, a firm or a customer, your data contains all of the important tales, which, when expressed clearly and elegantly, can be a great tool for data observation and interpretation. It is true that a picture is worth a thousand words if it is correctly composed, since an eye-catching graphic can more accurately clarify information. Mindmaps is a diagramming tool that is utilized in the development and visualization of structure and relationships, classification of ideas, observation and organization of information, decision-making, problem-solving, and writing, and general work morale enhancement. Different approaches are used in respect to the scope, such as:

- Visual Information - This is most commonly employed in news media, where information on current events is given in the form of a tree map in print, television, and the Internet, among other places, in order to reach a large audience.
- Data Visualization - In today's environment, it's nearly difficult for a corporation to make a strategic decision without consulting an accurate figure.
- Displaying connections - It establishes a link between variables such as story, relevance, and usability, all of which are related to one another.
- Displaying Websites - It arranges all of the search results in a three-dimensional format.

D. Application of data visualisation

The bulk of visualization designs serve as tools for cognitive enhancement and are meant to boost decision-making. It's crucial to consider the intended application of the visualization when designing and creating a data visualization prototype. Data visualization involves choosing and reevaluating the numbers that were used to produce the representation in addition to just presenting them.

Data visualization [2] is a crucial branch of computer science with a wide range of applications. In several domains of health and science, application-specific tools have been developed to assess individual datasets.[3]

Public health: The success of public health surveillance depends on the ability to evaluate and display data in a clear and understandable manner. Health researchers desire dependable and advanced technologies to improve their work. Security is essential in cloud-based medical data visualizations. Today, a wide range of graphical representations can be found in any medical or health magazine.

Environmental Science: Environmental managers need visualization since they must base their decisions on incredibly complex data. Applications for visualization in applied environmental research are beginning to take shape. Data visualization is essential in the early phases of a fraud investigation. Fraud investigators might utilize data visualization as a proactive detection technique to identify patterns that point to fraudulent behavior.

E. Challenges in data visualisation

Because of the volume, variety, and velocity of data, big data visualization is difficult. Navigating enormous data quantities and effectively displaying useful and usable findings of data visualization and analysis are the most challenging aspects of working with big data. In order to help decision makers grasp data in an easy and direct way utilizing graphs and charts, new ways to study data must be devised. Extremely large data sets cannot be handled by traditional visualization approaches. We should be able to receive graphics from the visualization tool with the least amount of latency possible. Parallelization is also essential to deal with such a large volume of data, which is a hurdle in visualization. Big data visualization's main purpose is to find intriguing patterns in large amounts of data.

Big data, both organized and unstructured, creates a new set of challenges for the production of visualizations. This is due to the fact that we have to take the speed, volume, and diversity of the data into account. Performance, operability, and degree of discrimination-related issues are posing new challenges for large data visualization and analysis. A large simulated data collection requires a lot of work and time to create. It's also challenging to figure out which graphic is the greatest to employ.

III. ANALYSING COVID-19 DATA

Big data, both structured and unstructured, also presents a new set of issues for visualisation development. The purpose of this study is to analyze covid-19[4] pandemic data. The data that has been used for this study is from the start of the pandemic i.e., Jan 2020 to August 2021. Here, we will be studying the relationship between the death and the confirmed cases of covid-19 and develop a model by using the same data. Once the model will be ready, we'll use it to predict death based on the number of confirmed cases per day. Regression techniques will be used for this purpose. Before any data visualization techniques are applied on the data set it must be pre-processed and cleaned. For all the operations performed on the data set, we will be using the R programming language. The following are the steps taken.

Step 1: Install various packages to be used in the whole process, these include "janitor", "ggplot2", "lubridate", "dplyr" and "tidyverse".

Step 2: The data contains the following columns, sno, Date, Time, State/Union Territory, Confirmed Indian Nationals, Confirmed Foreign Nationals, Cured, Deaths, Confirmed and year. First read the data into the system and save it as a variable, say covid.data using read.csv() function.

There are a few column heads that have space in them but to use them as a variable they shouldn't have any spacing

between them. To clean the column names, use `clean_names(covid.data)`

Step 3: There are a few columns that wouldn't be required for our study so we need to eliminate them, these are Confirmed Indian Nationals and Confirmed Foreign Nationals. Also, remove empty rows and columns and convert the date column data into date format using `as.Date()` function. And change the state values to factors using `as.Factor()` function. The data is cleaned for our purpose.

Step 4: Since the data is available for all the days for the given duration, it can be represented through a plot using the `ggplot()` function.

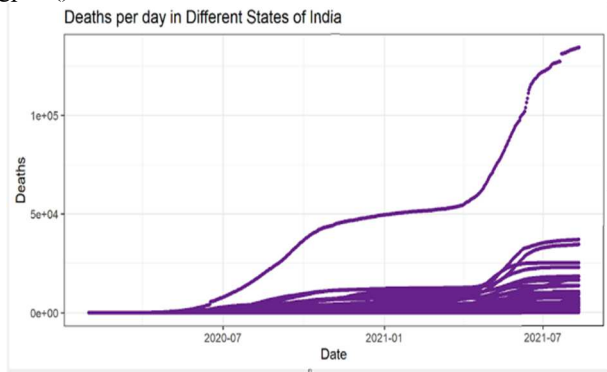


Fig 2. Deaths per day in India

Figure 2 represents the rising death cases in India for different states. Because one common color is used for all the states there is no clarity on which state cases are rising faster. However, it is clear that one of the states has the greatest number of rising cases. From the following plot a conclusion that can be made is during the first half of 2020 there were a smaller number of deaths, it increased in the second half of 2020 and a major increase was seen in the first half of 2021. It is clear from this graph that most deaths per day in the first wave were in September. Again, the number of cases started increasing with the second wave from May 2021.

However, data for different states can be studied separately. We will be studying data of Andhra Pradesh for the same duration from this dataset only. Other state data can also be studied. Figure 3 depicts the same nature as the plot for the whole country. It is showing a similar increase in deaths cases for the first and second wave of Covid-19.

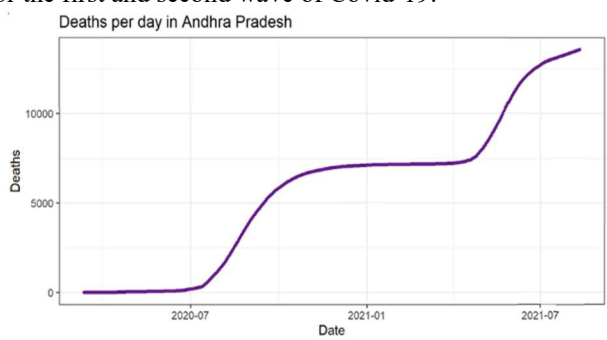


Fig 3. Daily deaths in Andhra Pradesh

Step 5: This data can be summarized for the two years separately

From Fig. 4 it is clear that the deaths increased at a higher rate in 2021 from the very start. Figure 5 shows the same behavior as Fig 3 for the year 2020.

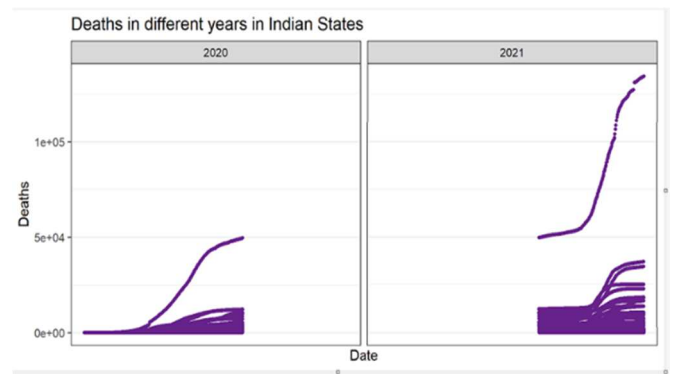


Fig 4. Deaths in 2020 and 2021 in India

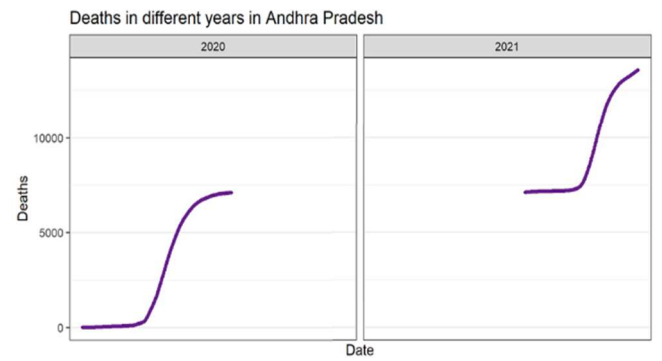


Fig 5. Deaths in 2020 and 2021 in Andhra Pradesh

Step 6: The number of deaths has been represented above. But to find the relationship between confirmed cases[5] and deaths, confirmed cases should also be plotted.

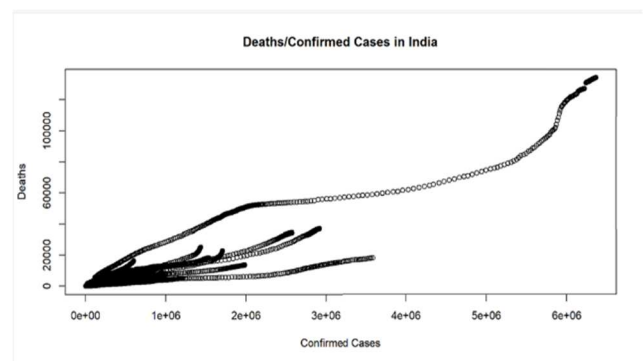


Fig 6. Deaths and Confirmed Cases Plot in India

The graph in Fig. 6 shows that with the increase in the number of cases, the number of deaths also increased. A similar behaviour can be observed [6] in a graph for Andhra Pradesh as shown in Fig. 7.

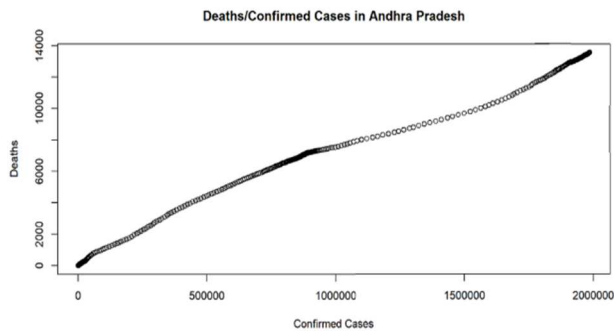


Fig 7. Death and Confirmed Cases plot in Andhra Pradesh

Step 7: The above study of the relationship between confirmed cases and deaths can be extended for the year.

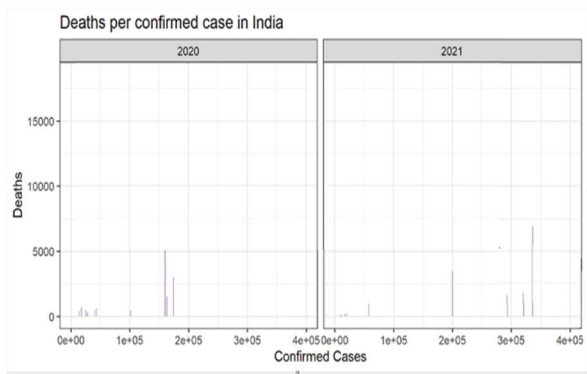


Fig 8. Death and Confirmed Cases in India

As shown in Fig.8, it can be observed [7] that the number of deaths increased with the increasing number of death cases in India. Similarly, as shown in Fig. 9 the data for Andhra Pradesh for 2020, it can be observed [8] the number of deaths were directly related to the number of confirmed cases.

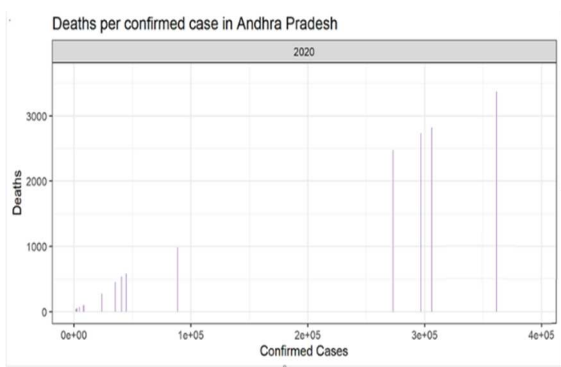


Fig 9. Death and Confirmed Cases in Andhra Pradesh

Step 8: This data can be plotted with the help of dots for monthly data. But monthly data is not available readily for analysis. So a new column for month should be added to the dataset using the mutate() function and the following code:

```
covid.Data <- covid.Data %>%
mutate(month = month(date))
```

Here, month() function helps in extracting month from the respective date. This will add a new column in the dataset. Once the month has been extracted, a new dataset can be

made which contains the total number of deaths in the complete month, i.e. the sum of cases of all the dates of that month. The code for creating new dataset named covid.Data_month is:

```
covid.Data_month <- covid.Data %>%
group_by(month) %>%
summarise(sum_deaths = sum(deaths))
```

We can now use this to plot data of deaths per month.

Step 9: The monthly number of deaths for India can be plotted as shown in Fig 10. It can be observed from Fig. 10, that the number of deaths reached its peak in the seventh month and then decreased again. The monthly number of deaths data for Andhra Pradesh depicts the same behaviour of the number of deaths as shown in Fig. 11. Step 10: The month data can be represented through a bar plot as shown in Fig 12.

It can be observed from Fig 12. that there was an increase in the number of deaths from July 2020, it kept on increasing till July 2021, with a drop in February 2021. After August 2021 the cases again started dropping. The maximum number of deaths reported in India were in the month of July 2021. The bar plot of Andhra Pradesh shows a similar behaviour as shown by Fig 13.

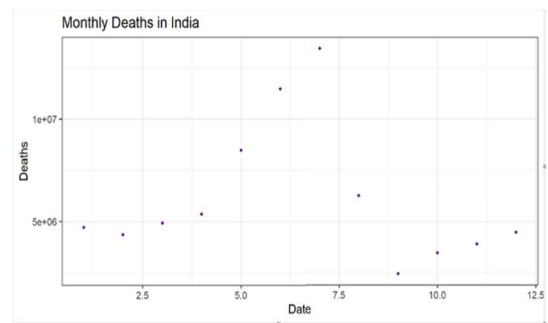


Fig 10. Monthly Deaths in India

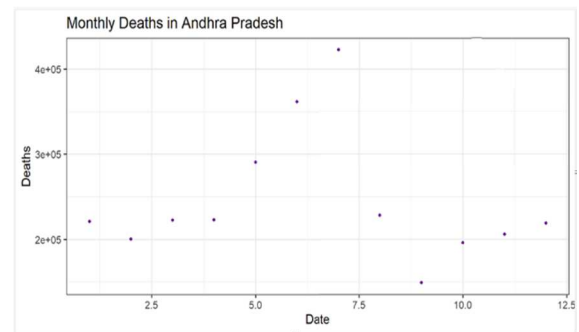


Fig 11. Monthly Deaths in Andhra Pradesh

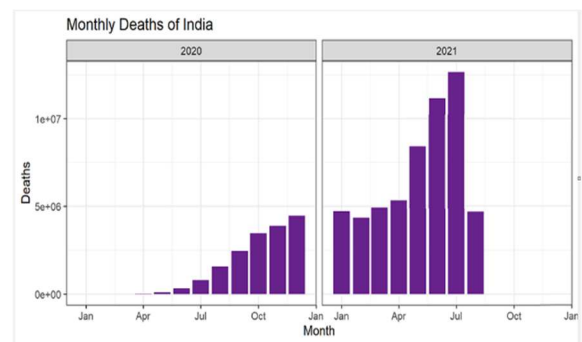


Fig 12. Monthly Deaths in India

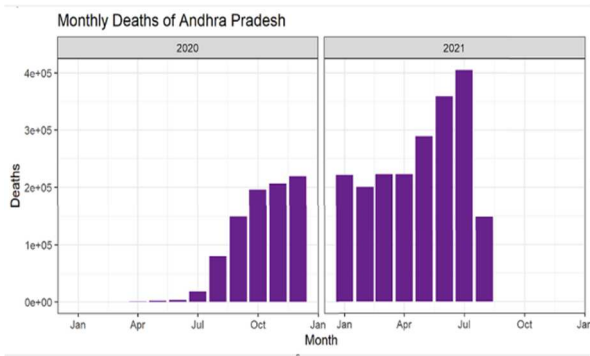


Fig 13. Monthly Deaths in Andhra Pradesh

Step 11: We have visualized the data and observed which months reported the maximum number of Covid-19 deaths[9] in India and Andhra Pradesh. Now, we find the correlation between the confirmed cases and deaths in India. It is 0.9180012.

Correlation ranges from -1 to 1. Here, -1 represents perfect negative correlation, 1 represents perfect positive correlation and 0 indicates no correlation. Since we have approximately 0.91, therefore the two variables are positively correlated. With increase in one, we can observe increase in other. The correlation between confirmed cases and deaths in Andhra Pradesh is 0.9908245, it is very close to perfect correlation.

```
Call:
lm(formula = deaths ~ confirmed, data = covid.Data)

Residuals:
    Min       1Q   Median       3Q      Max
-36170      17      514      555     37665

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.437e+02  3.541e+01  -15.36  <2e-16
confirmed    1.526e-02  4.906e-05   310.94  <2e-16

(Intercept) ***
confirmed ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4321 on 18044 degrees of freedom
Multiple R-squared:  0.8427,    Adjusted R-squared:  0.8427
F-statistic: 9.669e+04 on 1 and 18044 DF,  p-value: < 2.2e-16
```

Fig 14. Model for covid-19 confirmed cases and deaths in India

```
Call:
lm(formula = deaths ~ cured + confirmed, data = covid.Data)

Residuals:
    Min       1Q   Median       3Q      Max
-36388     -27      509      544     36654

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.305e+02  3.544e+01  -14.97  < 2e-16
cured        4.734e-03  7.786e-04   6.08  1.23e-09
confirmed    1.083e-02  7.296e-04   14.85  < 2e-16

(Intercept) ***
cured ***
confirmed ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4317 on 18043 degrees of freedom
Multiple R-squared:  0.843,    Adjusted R-squared:  0.843
F-statistic: 4.846e+04 on 2 and 18043 DF,  p-value: < 2.2e-16
```

Fig 15. Model for covid-19 confirmed, cured cases and deaths in India

Step 12: Now we shall train a model using the lm() function. First, we will train a model for finding relationships between

confirmed cases as independent and deaths as dependent variables. Then find the relationship between confirmed cases and cured as independent and deaths as dependent variables.

Once the model is trained, it can be summarized using summary() function [10].

```
covid.model <- lm(deaths ~ confirmed, data = covid.Data)
summary(covid.model)
```

The output is shown in Fig 14.

Similarly for two independent variables we have:

```
covid.detailedModel<-lm(deaths ~ cured + confirmed, data =
covid.Data)
summary(covid.detailedModel)
```

The output of the above code is in Fig. 15.

Step 13: Finally, the predict() function can be used to predict the number of deaths when a rise in the number of cases is seen in the country. This model has 89% accuracy.

IV. RESULT AND DISCUSSION

It was observed that the number of deaths increased in the first wave of covid-19 from July 2020. And in the second wave it increased from May 2021. The number of deaths were directly related with number of confirmed cases which had a correlation of approximately 0.91. A model was developed for this dataset which could be used further for prediction. The technique used for this purpose was regression, and the trained model gave 89% accuracy.

With a total size of 3.287 million km2, India is the world's second-most populous country, therefore assessing the spread of infection state-by-state would aid the government in making the best use of available resources. In terms of health-care facilities, India does not have the greatest in the world, and given the country's population, it is critical to use resources as efficiently and effectively as possible. Despite not having the best health-care facilities like other industrialised countries such as the United States, Italy, Germany, and many others, India has handled the pandemic well.

The further investigation involved the daily increase in the number of confirmed and death cases for one state and similarly, it can be extended to other states also. Regression models with good accuracy provides a solution for predicting data and analysing it further. Data visualization help in faster results and easier interpretations. Therefore, it is very popular in the many fields.

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